AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 1 · STUDENT LESSON

AI or Not?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

AI or Not? Follow the decision.

A red bus display, an umbrella forecast and a photo library all lead to one question: what produced this particular decision?

Keep one habit: Explain the decision → point to its evidence → state what remains unknown.

MAIN LESSON · SECTION 1

What makes a machine seem clever?

> Your goal: explain how a decision was made, using a fact about the process.

Start with one small puzzle

A clock says the correct time. Hearing it tells you what it did. It does not yet tell you how it did it. Keep that distinction in mind throughout this week.

A classroom clock speaks the time. Write one thing you observed and one thing you cannot determine just from hearing it.

By the end, you can...

- I can distinguish an observed result from an explanation of how it was produced.
- I can identify **written rule**s, **learned model**s and **human decision**s inside one product.
- I can explain what additional **evidence** is needed when a **mechanism** is unknown.

How to use this week

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

▼ **Earlier target wording and repeated recall**

Your learning target

Remember before reading

Look for **evidence** of how the system reasons or learns, not just its label or screen.



LEARNING TARGET A bus displays a delay **prediction** learned from past journeys. A **written rule** turns the display red above 10 minutes. A driver can override the message. Identify all three **mechanism**s.

Would seeing a red display prove that learning occurred?

Learner name:

Readiness for the connected explanation

A classroom clock speaks the time. Write one thing you observed and one thing you cannot determine just from hearing it.

This week’s end goal

By the end, I can separate a learned **prediction**, a **written rule** and a **human decision**.

MAIN LESSON · SECTION 2

A red bus display: what do we really know?

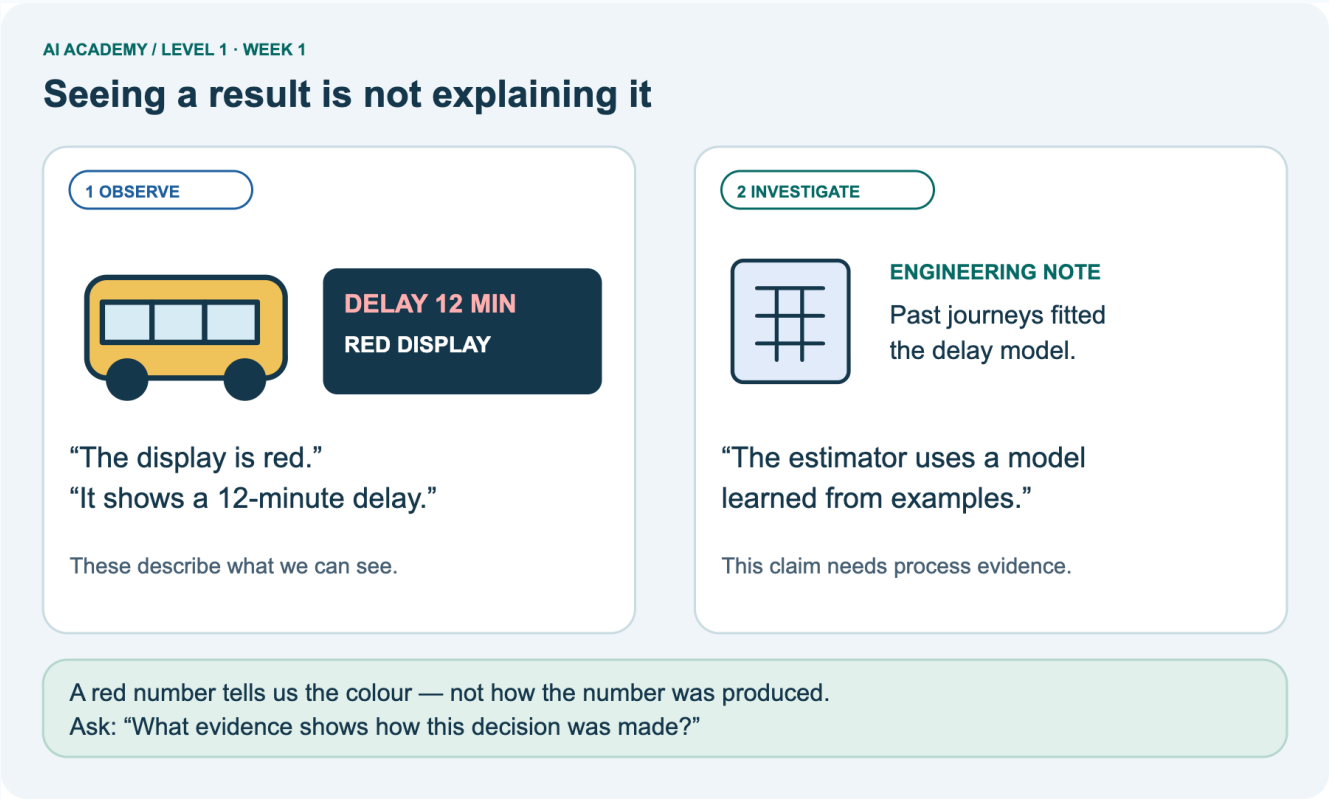
> First describe what you can see. Then look for evidence about what happened inside.

Meet Leila and Arun

The school bus display says DELAY **12 MINUTES** in red. Leila calls it AI because it predicted the delay. Arun calls it a simple rule because the number turned red.

Both have noticed a real part of the product, but neither has explained the whole system. To investigate, ask which decision you are talking about and what **evidence** reveals how that decision was made.

Look before you explain



Follow the labelled picture with the matching explanation. Open the vector diagram

An **output** is what a system produces. A **mechanism** is how it produces that **output**. Seeing a red number establishes its colour; it does not reveal where the number came from.

Similar **output**s can come from different **mechanism**s, and one product can contain several **mechanism**s.

Your first observations

- Name three visible details without explaining them yet.
- Circle the detail most connected to the system's decision.
- Name one detail that may be decoration or distraction.
- Make a first **prediction** and state the clue that supports it.

Read the builder's notes

These notes are given facts for our fictional case. They are stronger mechanism evidence than the colour of the display.

For this lesson we have an engineering note: the delay estimator was fitted using records of earlier bus journeys; a person wrote the colour instruction; and the driver can replace the message.

These are supplied facts for our classroom case. In a real investigation, we would need trustworthy documentation, a demonstration of the **process** or another source of **mechanism evidence**.

▼ **Extra observation prompts and original picture-story notes**

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A remote-control car, calculator, motion light, smart speaker, photo sorter, and robot vacuum may all look clever, but they do not make decisions in the same way.

What do you notice?

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

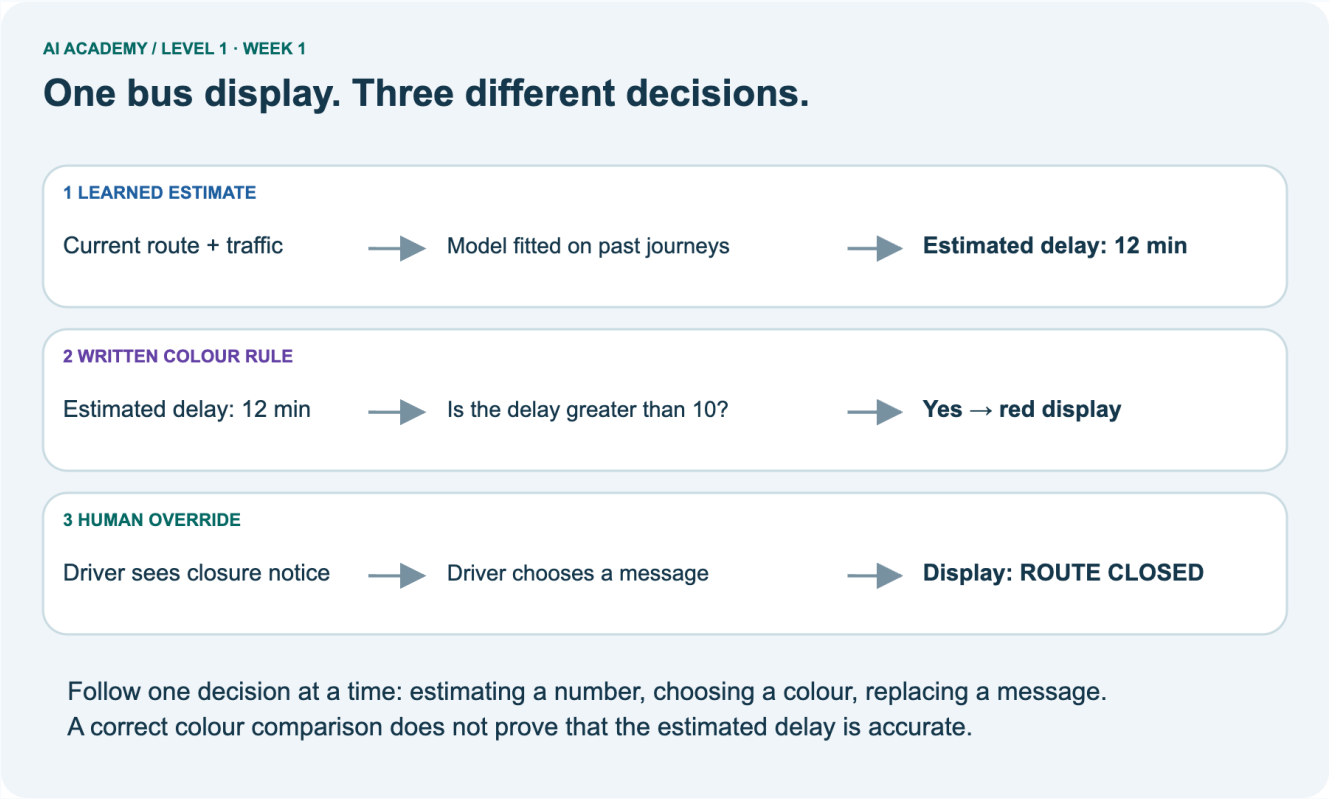
A closer explanation

MAIN LESSON · SECTION 3

One product, three different decisions

> Follow the number, the colour and the driver's message separately.

Read the whole system



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain each decision

1. Predict a number

Earlier journeys were used to fit the delay model. It uses today's route and traffic information to estimate a 12-minute delay. A prediction can be wrong.

2. Apply a written rule

A person wrote this instruction: turn the display red when the estimated delay is greater than 10 minutes. Twelve is greater than ten, so the display turns red.

3. Make a human choice

The driver can choose a replacement message, such as ROUTE CLOSED. That replacement is a human decision.

Trace an information path

1. **INPUT** a command, signal, photo, sound, or number
2. **PROCESS** a human choice, fixed instruction, or learned pattern
3. **OUTPUT** movement, calculation, action, or **prediction**

Keep this important distinction

AI can also reason using knowledge and rules. So "written rule" does not always mean "outside AI." In this bus case, changing the colour is a simple automatic step. Classify each decision separately.

Explain one arrow

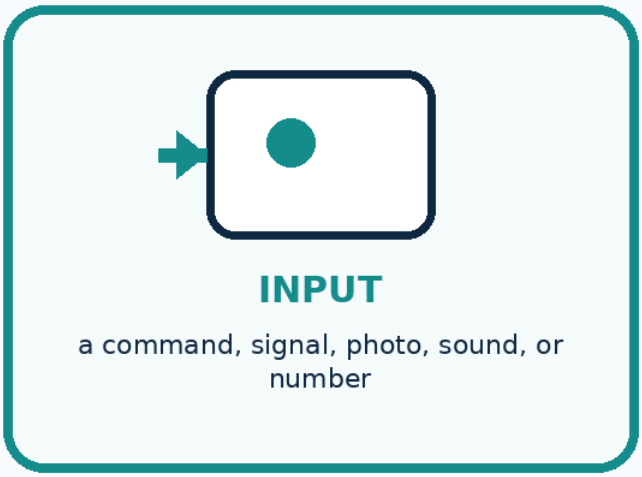
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

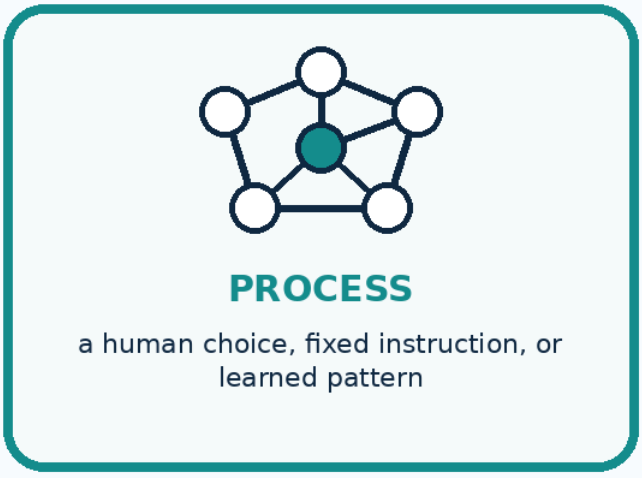
▼ Full bus table and further mechanism explanation

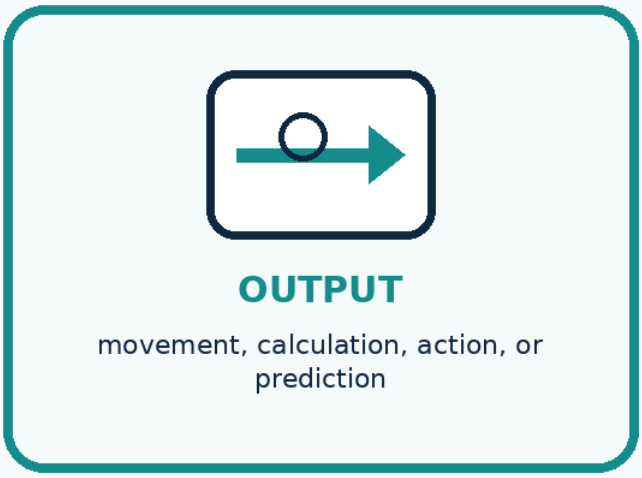
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







CORE IDEA Look for **evidence** of how the system reasons or learns, not just its label or screen.

UNDER THE HOOD A system should be classified by its decision **mechanism**. **Human control**, **fixed automation**, and model-based **prediction** can produce similar visible behavior, so appearance and marketing words are weak **evidence**.

Draw a fresh example with all three stages:

A closer explanation

The bus display contains three mechanisms

INFORMATION ENTERING	HOW THE DECISION IS MADE	RESULT
Current route and traffic	Delay model fitted on earlier journeys	Estimated delay of 12 minutes
The estimated delay	Written rule red when delay is greater than 10	Red display
Driver reads a closure notice	Driver chooses an override message	Display says route closed

Read each row from left to right. The first row predicts a number, the second chooses a colour, and the third shows a human override. A correct colour does not establish that the delay prediction is correct.

In the first row, **fitting** means adjusting a model using examples. Applying the fitted model to a new journey is a different step. The model can produce an incorrect estimate even if it was trained carefully.

In the second row, a written threshold compares the estimate with 10. It can follow its instructions perfectly while using a bad estimate. In the third row, a person makes the message decision.

Connect the picture and the explanation

AI is a broader field than learning from examples. It also includes approaches that represent knowledge and reason with rules. Therefore, do not use the shortcut that every rule is non-AI.

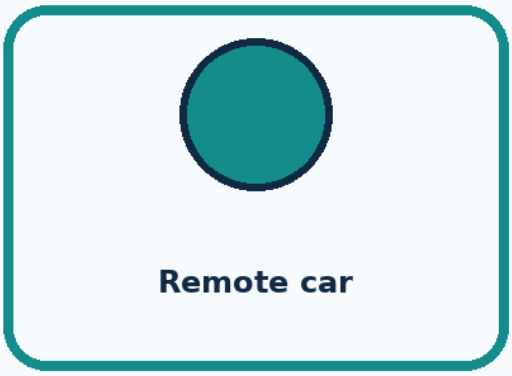
In this case, a simple colour switch is ordinary automation; identifying that switch does not classify every other part of the bus system.

MAIN LESSON · SECTION 4

Compare the evidence, not the appearance

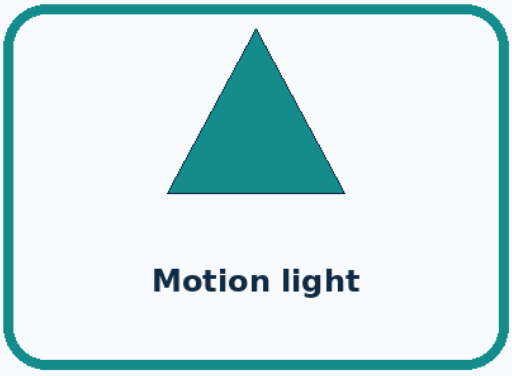
> Ask who or what chooses the action in each case.

Four mechanisms to compare



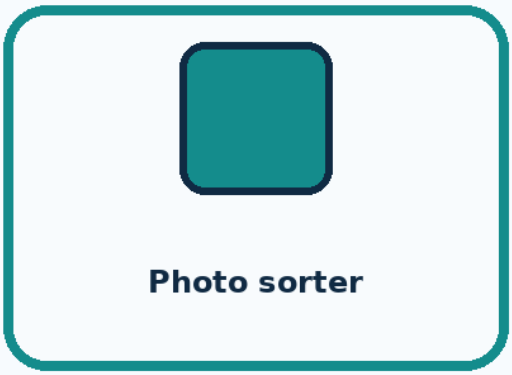
The case to investigate

CASE 1 — Remote car Clue: person sends each steering command. Meaning: human-controlled.



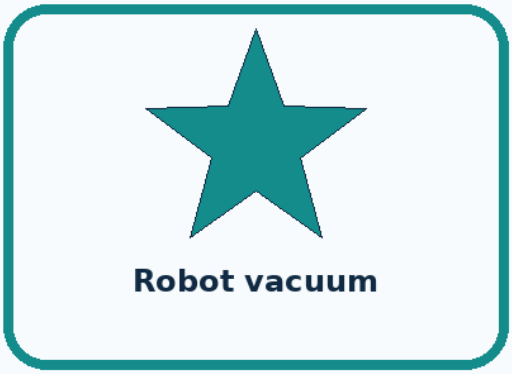
The case to investigate

CASE 2 — Motion light Clue: written if-motion-then-light rule. Meaning: **fixed automation**.



The case to investigate

CASE 3 — Photo sorter Clue: trained model predicts a new label. Meaning: AI-based.



The case to investigate

CASE 4 — Robot vacuum Clue: description hides the **mechanism**. Meaning: **not enough information**.

Discuss with a partner

- Which two pictures are easiest to confuse? What **evidence** separates them?

A limit to remember

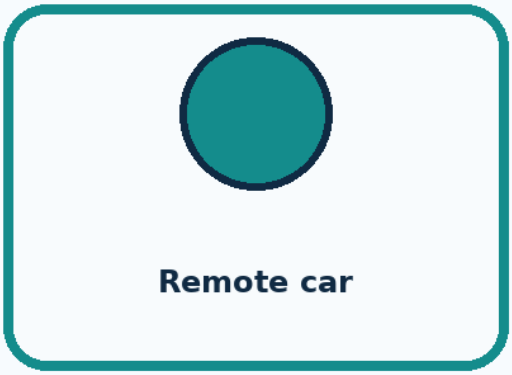
BOUNDARY Do not call a system AI because it looks, moves, or talks like a person.

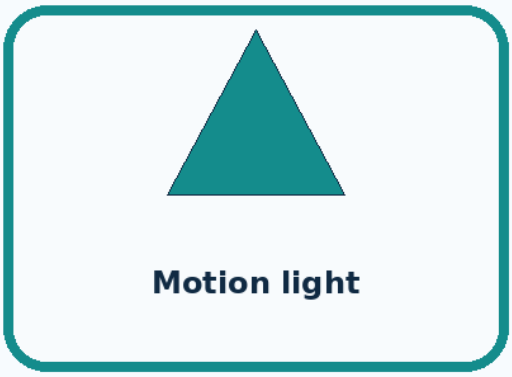
▼ **Original comparison cards and umbrella extension**

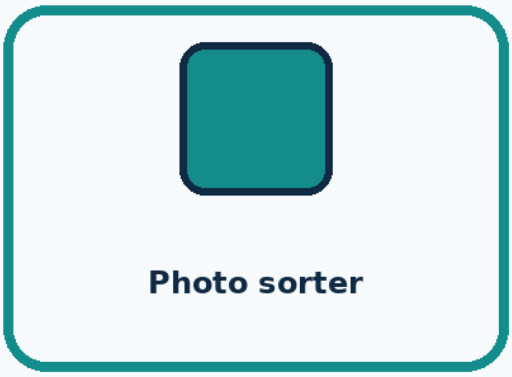
PICTURE GALLERY

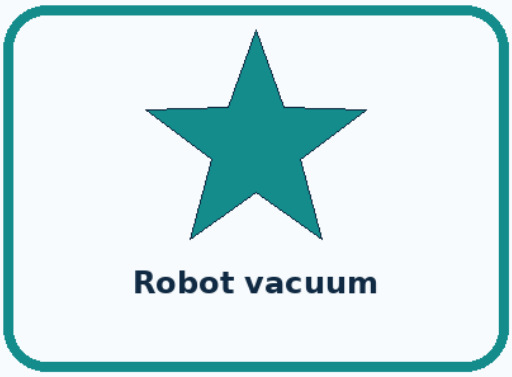
Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Remote car Clue: person sends each steering command. Meaning: human-controlled.

CASE 2 — Motion light Clue: written if-motion-then-light rule. Meaning: **fixed automation**.

CASE 3 — Photo sorter Clue: trained model predicts a new label. Meaning: AI-based.

CASE 4 — Robot vacuum Clue: description hides the **mechanism**. Meaning: **not enough information**.

Change one thing in the pictured case

Change: Keep the forecast but raise the display boundary to **80%**.

Prediction: The display no longer says TAKE ONE.

Reason: 70 is below 80; changing a display rule does not refit the model.

MAIN LESSON · SECTION 6

Work through the umbrella display

> The forecast and the display rule do different jobs. Change one while keeping the other fixed.

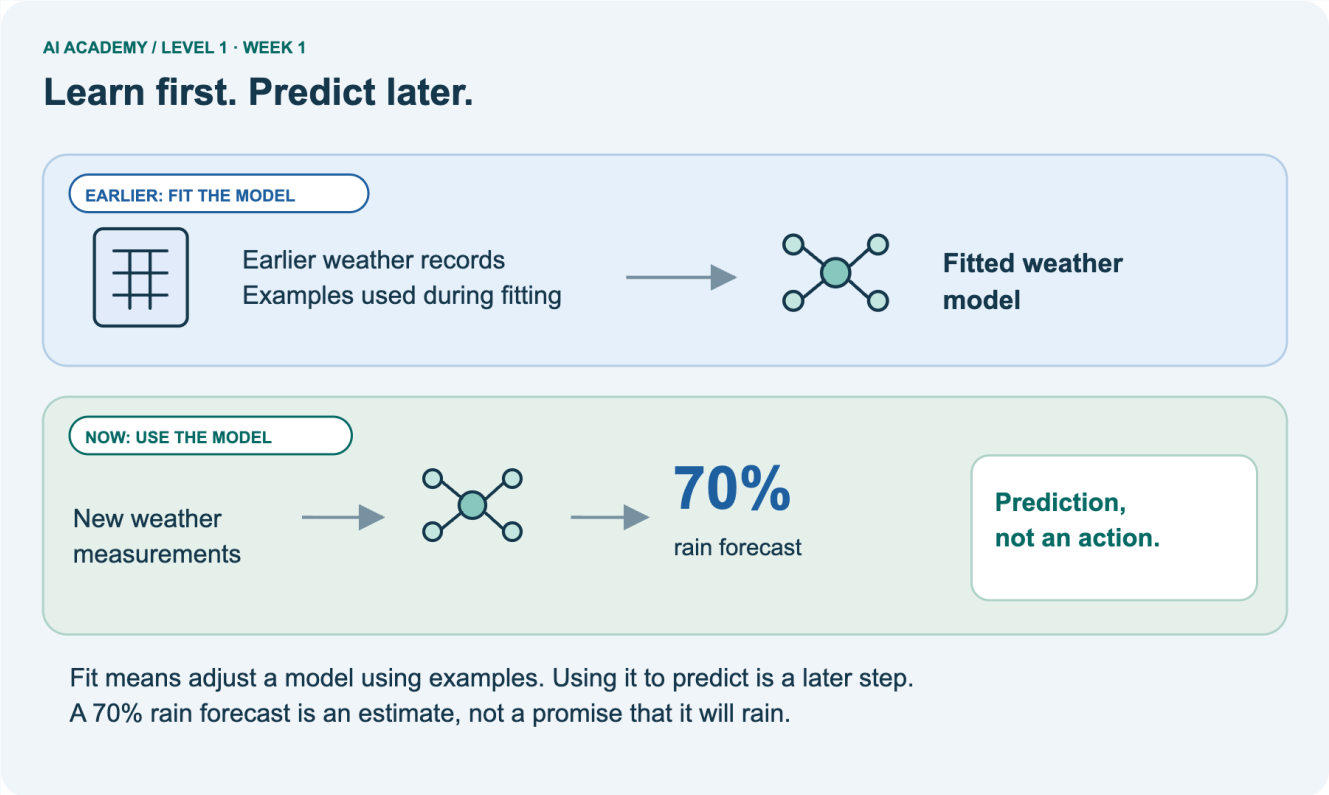
The case and its given facts

A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?

A weather model was fitted on earlier weather records; its forecast is **70%** rain.

A **written rule** shows TAKE ONE at rain forecasts of **60%** or more. The teacher decides whether to distribute umbrellas.

First learn; later predict



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the four stages

AI ACADEMY / AI-1/1

AI or Not?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Weather records

Earlier observations were used to fit the forecasting model.



2. Forecast

The learned model outputs 70% rain, not a guaranteed shower.



3. Display rule

70 is at least 60, so the written rule displays TAKE ONE.



4. Final decision

The teacher checks current conditions before distributing umbrellas.

CHANGE ONE THING

Keep the forecast but raise the display boundary to 80%.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Weather records	Earlier observations were used to fit the forecasting model.
2. Forecast	The learned model outputs 70% rain, not a guaranteed shower.
3. Display rule	70 is at least 60, so the written rule displays TAKE ONE.
4. Final decision	The teacher checks current conditions before distributing umbrellas.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why the answer follows

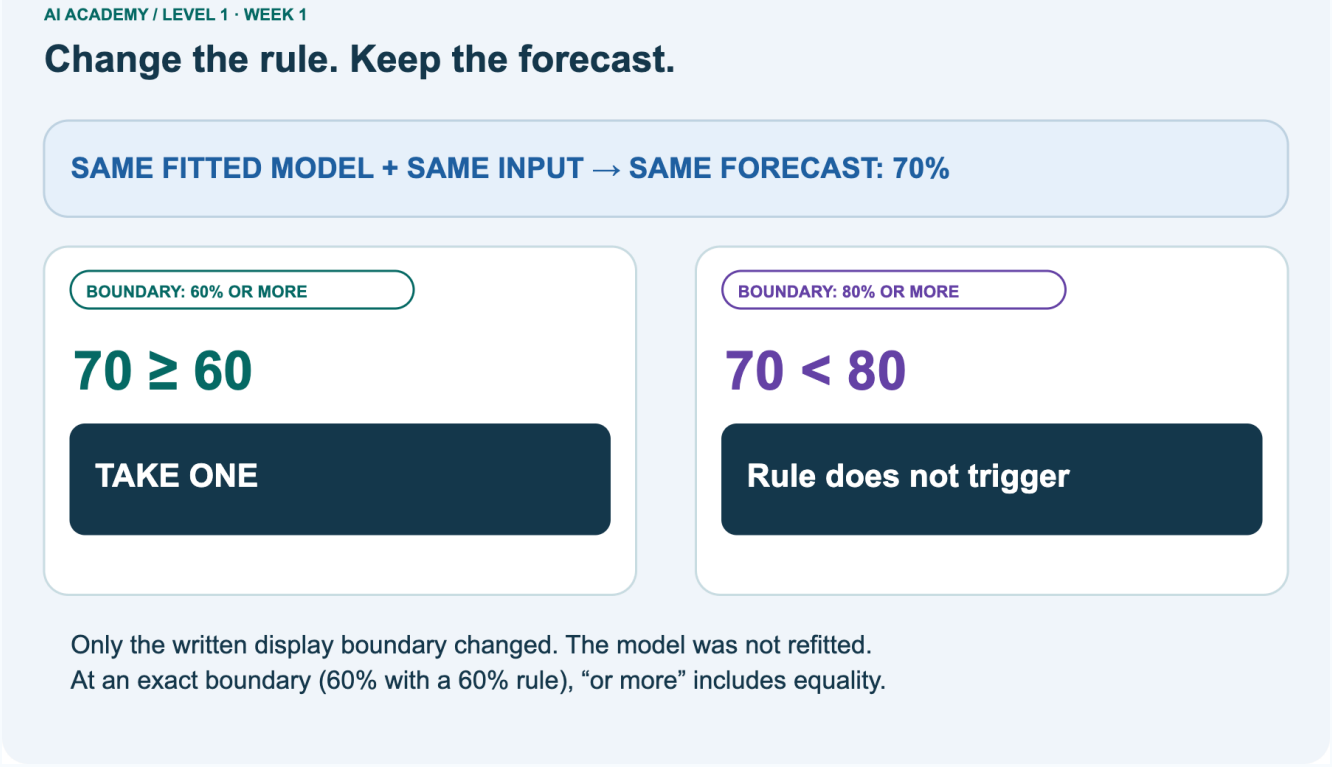
The **fitting** record supports calling the forecasting stage learned. Comparing 70 with 60 is an explicit rule. The final action belongs to a person.

Merely seeing TAKE ONE would not reveal any of these **mechanism**s; the **process** notes supply that **evidence**.

Predict: keep 70%, change the boundary

- Change one thing: Keep the forecast but raise the display boundary to **80%**. Predict the result before checking the explanation. What remains the same?

▼ Check the changed-rule explanation



Follow the labelled picture with the matching explanation. Open the vector diagram

What this cannot prove

The display can follow its written rule correctly even if the forecast is wrong. Knowing why TAKE ONE appeared does not prove that rain will happen.

Explain the deciding step

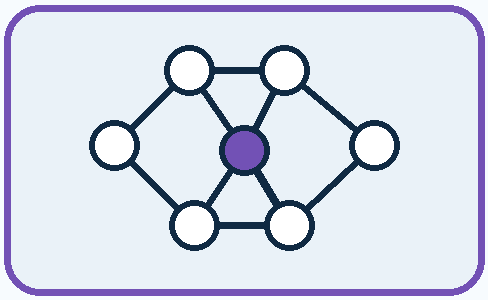
- Read the labelled picture: Point to the learned stage. Does the changed display prove the weather forecast changed?

▼ Bus reasoning recap and earlier worked-example prompts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A remote-control car, calculator, motion light, smart speaker, photo sorter, and robot vacuum may all look clever, but they do not make decisions in the same way.

STEP 1 — OBSERVE a command, signal, photo, sound, or number

STEP 2 — TRACE a human choice, fixed instruction, or learned pattern

STEP 3 — CONCLUDE movement, calculation, action, or **prediction**

MODEL REASONING Follow the information and identify who or what makes the important choice. A robot body, screen, voice, or the word smart cannot prove AI.

If two **mechanism**s fit the **evidence**, say that **more information** is needed.

DO NOT OVERCLAIM Do not call a system AI because it looks, moves, or talks like a person.

What new **evidence** could change this answer?

A closer explanation

Question: Does the red display prove that a **learned model** was used? First separate the claim from the **observation**. The **observation** is that the displayed number is red. The claim concerns the **process** that produced the number.

Next find relevant **evidence**. The engineering note says earlier journeys were used to fit the delay estimator. That note supports a learned-model classification for the estimator. The red colour alone does not.

Claim	Relevant evidence	Conclusion
The estimator uses a learned model	The note describes fitting on past journeys	Supported for the estimator
The red colour proves learning	Only the displayed colour is observed	Not supported by colour alone
The driver override is a model prediction	A driver selects the replacement message	Contradicted in this case

A strong answer states its limit: I can identify these three **mechanism**s from the supplied note, but I cannot infer the model's accuracy from one estimate.

Even if the bus arrives exactly **12 minutes** late, one success does not establish that the estimator will always work.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: AI or Not?

The facts we will use

Why this result follows

What this example does not establish

A correct rule comparison does not establish that rain will occur or that **70%** forecasts are followed by rain on about 70 out of 100 comparable days.

Repair a tempting explanation

> A robot body, a screen or a correct answer does not reveal the decision mechanism.

Try two repairs

- CLAIM A Every robot uses AI.
- CLAIM B Anything with a screen is AI.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

What if the mechanism is unknown?

A robot vacuum is advertised as smart. The advertisement tells you almost nothing about its decisions. A useful question is not Does it look intelligent?

Ask instead how it selects a route, whether adjustable values were fitted using examples, or when a person supplies a command. It may combine several answers.

Sometimes the honest conclusion is that the **mechanism** is unknown. That is a precise answer when you identify the missing **evidence**. It is weaker to guess a **mechanism** and then treat your guess as a fact.

Keep **observation**s, supplied **evidence** and assumptions separate in your investigation.

Explain without guessing

> **A device speaks a correct answer. Explain why neither speech nor correctness alone identifies its mechanism**

▼ Comparison table and earlier repair prompts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Remote car	person sends each steering command	human-controlled
Motion light	written if-motion-then-light rule	fixed automation
Photo sorter	trained model predicts a new label	AI-based
Robot vacuum	description hides the mechanism	not enough information

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Your turn: explain a photo library

> Practise with help, then explain the evidence in your own words.

Read the case

A photo library predicts whether a picture contains a bicycle using a model fitted to labelled pictures. A **written rule** places all bicycle **prediction**s in one album. The owner can move any photo to a different album.

Complete the explanation in your workbook: the **prediction** stage is ___ because ___; the automatic album step is ___ because ___; the manual move is ___.

Complete the explanation

1. Use three rows: prediction → album placement → owner move. For each row, write who or what made that decision and the fact that tells you. If a photo is misplaced, check the intermediate prediction before deciding where the error occurred.

> **Complete the three classifications and name one record that would help diagnose a misplaced photo**

Investigate a wrong album

If a photograph lands in the wrong album, do not immediately blame the written album rule. The **prediction** might have been wrong, the album rule might have been implemented incorrectly, or a person might have moved the photo.

Inspect the intermediate **prediction** and the action record to distinguish these causes.

Challenge one generalisation

- Use the worked case for **Evidence** and decision **mechanism**s. Proposed explanation: "All rules are outside AI". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your explanation with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

▼ Umbrella practice and additional guided prompts

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Look for **evidence** of how the system reasons or learns, not just its label or screen.

Use a fresh example to explain AI system.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Anything with a screen is AI.

Invent one edge case and name the safe response.

A closer explanation

Practise with less help: Complete the photo library explanation

Use workbook W2 for the connected example "**Evidence** and decision **mechanism**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Check your reasoning

Explain the picture

Point to the learned stage. Does the changed display prove the weather forecast changed?

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

This is supported practice. Keep the separately named independent case for an attempt before feedback.

MAIN LESSON · SECTION 9

Investigate: become a device detective

> Use the supplied cards. Record what you predict, what the given facts support, and what is still unknown.

Your case cards

Use these six fictional cards: a lamp triggered by a written motion rule; a photo sorter fitted on labelled pictures; a car steered by a person; a calculator following arithmetic instructions; a smart speaker with no process notes; and a robot vacuum with no process notes. For the last two, the mechanism has deliberately not been supplied.

Run the investigation

1. State your **prediction** before the result is known.
2. Complete the task: Sort six devices by **mechanism** and defend two borderline cases.
3. Keep a normal case and an edge case clearly separated.
4. Record what happened—even when it does not match your **prediction**.
5. Explain the result, one limitation, and the next useful test.

Keep your evidence separate

Case	Prediction	Observed result	What it means
Normal			
Edge			

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

After you have tried

Use the six-device worked explanation in the reference bank to discuss your result. A discussion case is practice, not a fresh assessment.

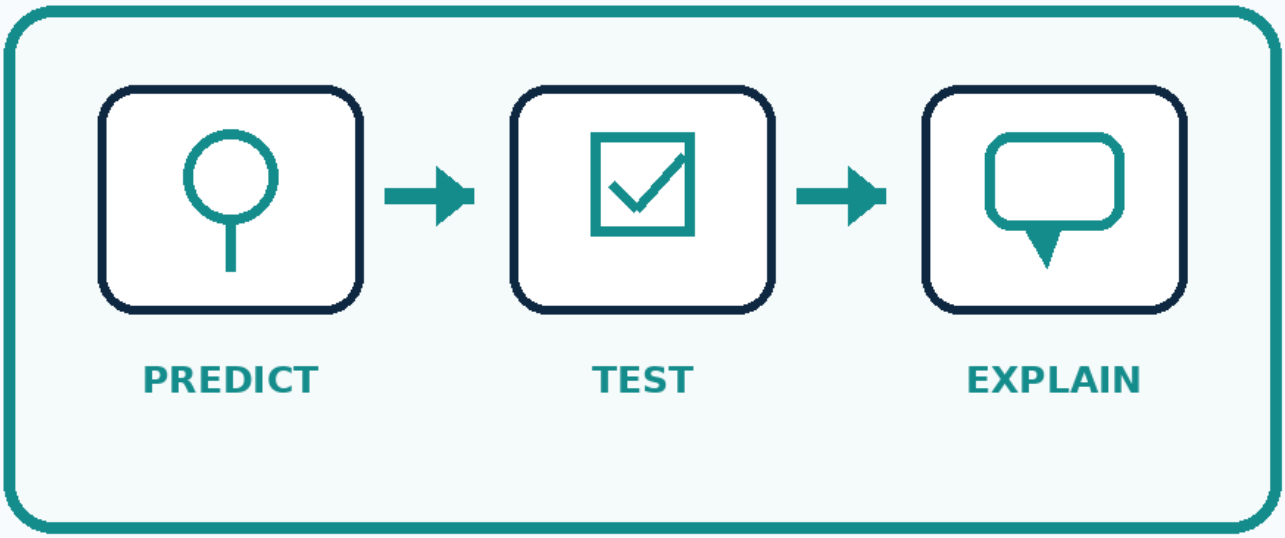
INDEPENDENCE After support is removed, repeat the proof with a fresh case.

▼ **Original lab illustration and headings**

EXPLORER LAB

Device Detective Sort

Predict first, test honestly, and explain what the **evidence** means.



Plan → try → check

MAIN LESSON · SECTION 14

Explain a fresh case on your own

> Use the greenhouse facts. Give one first attempt before hints or feedback.

The greenhouse challenge

A school greenhouse estimates tomorrow's temperature using a model fitted on historical weather. A **written rule** displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent.

Identify each **mechanism** and explain why a correct CHECK PLANTS message would not prove an accurate temperature **prediction**.

My independent explanation, trace and limits

Record your support

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Other transfer questions for later practice

Independent case: Investigate a new hybrid system

Week 1 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A toy follows your steering buttons. A photo app learned from labelled pictures. A talking clock follows a written timetable. Which uses a **learned model**? What **evidence** supports each classification?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Choose whether the Helpful Classroom Sorter needs a person, fixed rule, or **learned model**; justify the smallest safe solution.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A bus displays a delay **prediction** learned from past journeys. A **written rule** turns the display red above 10 minutes. A driver can override the message. Identify all three **mechanism**s.

Would seeing a red display prove that learning occurred?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Start your Helpful Classroom Sorter

> Choose one real need, then justify the simplest process that could meet it.

Make one design decision

THIS WEEK'S CHECKPOINT Choose whether the Helpful Classroom Sorter needs a person, fixed rule, or **learned model**; justify the smallest safe solution.

For the Helpful Classroom Sorter, annotate which decisions follow written instructions and which, if any, were fitted from examples.

Plan your evidence

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Keep the boundary clear

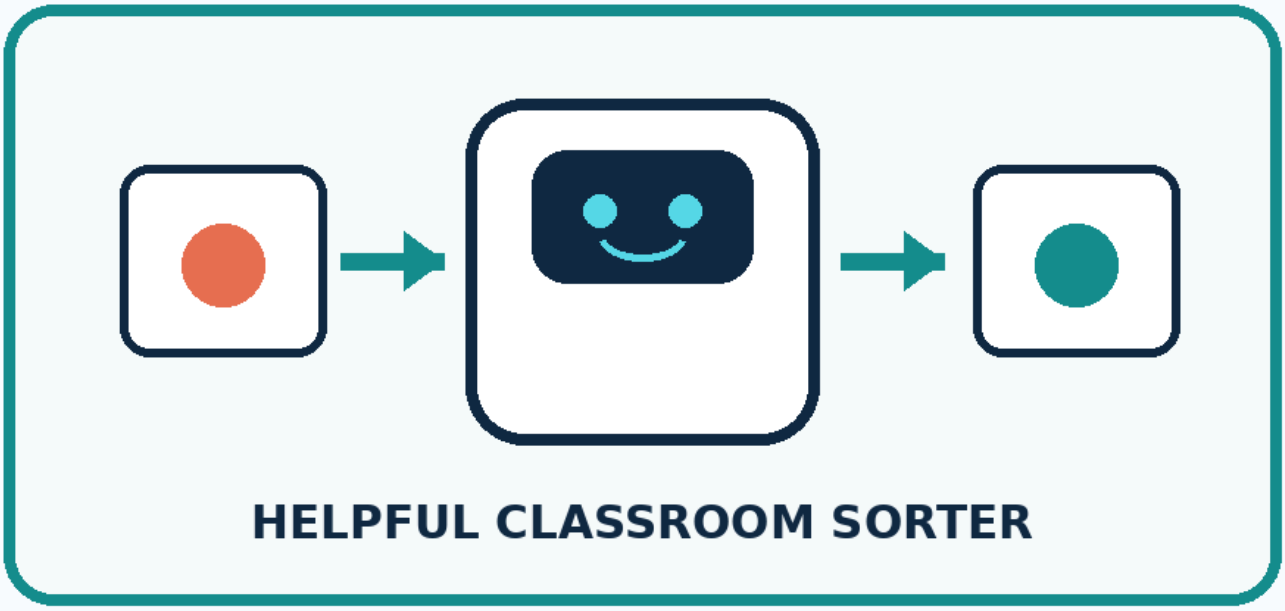
RESPONSIBLE DESIGN Do not call a system AI because it looks, moves, or talks like a person.

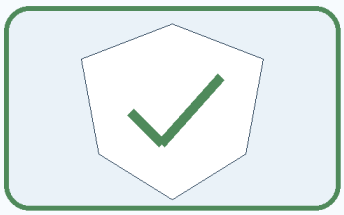
▼ **Project illustration, original heading and record label**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish with one explanation you can defend

> Close your notes, explain the idea, then decide what evidence to keep.

Say it without the page

- Close your notes. Explain this goal in your own words: Distinguish an observed result from an explanation of how it was produced

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Choose your next move

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Return here when a word blocks your explanation; you do not need to memorise both tables.

Useful words in this week's cases

Word	Meaning
Mechanism	The process that produces a result
Written rule	An instruction specified by a person before the system applies it
Learned model	A system whose adjustable values were fitted using examples
Hybrid system	A product that combines more than one kind of process

Word	Meaning	My example
AI system	a system using learned patterns or symbolic reasoning	
fixed rule	an instruction written in advance	
human-controlled	a person makes the important choice	
evidence	a clue that supports or changes a claim	

Use a word to repair a claim

- CLAIM TO REPAIR Every robot uses AI.
- Use one vocabulary word and one piece of **evidence** to repair it:

<details>
<summary>Vocabulary badges and original teaching notes</summary>

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.





</details>





Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ **Choose a harder thinking challenge**

REFERENCE · SECTION 10

Choose a harder thinking challenge

> Use one challenge when you are ready; these are options rather than six more compulsory tasks.

Choose your next challenge

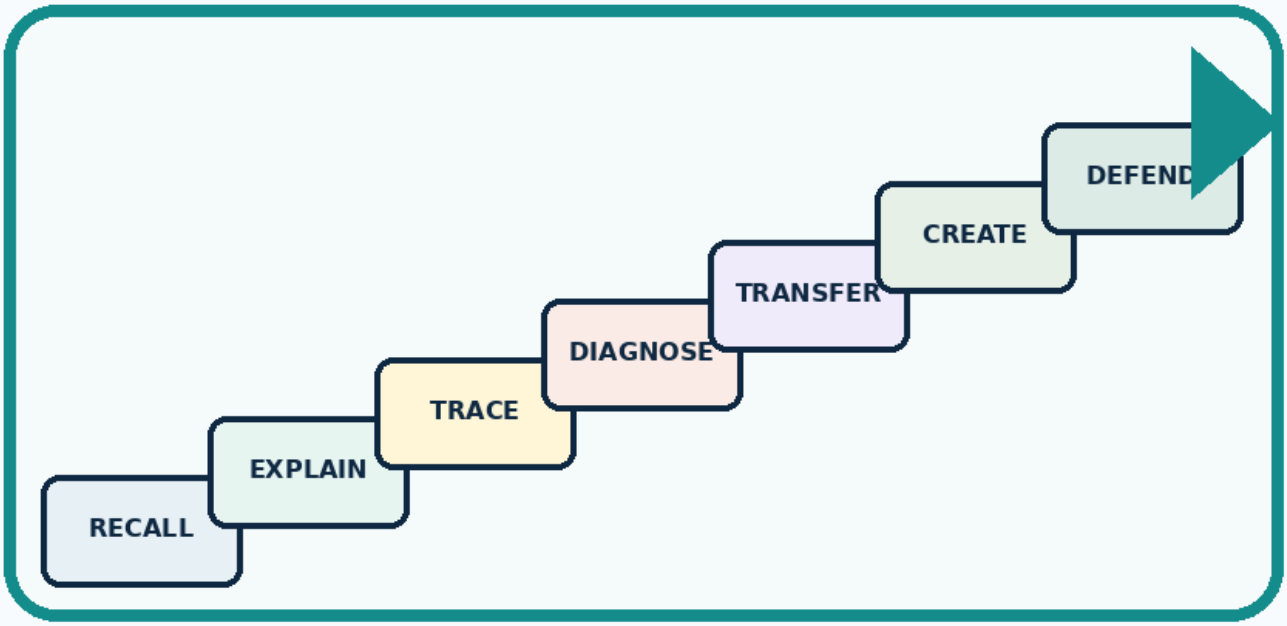
- RECALL Define AI system.
- EXPLAIN Look for **evidence** of how the system reasons or learns, not just its label or screen.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: Every robot uses AI.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Practice-ladder illustration and coaching note

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ A checklist for explaining any new case

REFERENCE · SECTION 12

A checklist for explaining any new case

> Use this as a reference after practice. The greenhouse is this week's selected independent case.

Check an explanation

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Think about your support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Original six-device assessment wording and extra notes

MASTERY

A Checklist for Explaining a Case

Use this checklist after practice. Complete the greenhouse case separately before feedback.



PRACTICE RECORD Classify the six supplied devices by **mechanism**, then defend two difficult cases with **evidence**. These devices have already been used in teaching; do not count the same cases as a fresh assessment.

Independent **evidence**:

SELF-CHECK Follow the information and identify who or what makes the important choice. A robot body, screen, voice, or the word smart cannot prove AI. If two **mechanism**s fit the **evidence**, say that **more information** is needed.

NEXT BRIDGE Carry today's **evidence** habit into: **Input**, **Process**, **Output**.

▼ **Check the six-device investigation**

REFERENCE · SECTION 13

Check the six-device investigation

> Open this after attempting the device detective task. It is a worked explanation.

Question and explanation

Six devices are a motion-triggered lamp, a trained photo sorter, a remote-controlled car, a calculator, an unspecified smart speaker and an unspecified robot vacuum.

Classify all six using **mechanism evidence**. Which two need **more information**?

The lamp uses a fixed rule; the photo sorter uses a **learned model**; the car follows human commands; the calculator performs arithmetic. The speaker and vacuum need **mechanism** details. Products can combine **mechanism**s.

A robot body or smart label proves none of them.

What the explanation does and does not establish

Scope note: this course mainly studies learning from examples. AI also includes symbolic and rule-based approaches. Ask how a system reasons or learns; a robot body or a simple automatic switch is insufficient **evidence**.

Explain your reasoning

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ **Repeated six-device table and wording**

Week 1 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Find the decision maker

Evidence or stage	Value or result
Motion lamp	Written rule
Photo sorter	Learned from examples
Remote car	Human command
Smart speaker	Need more information

Six devices are a motion-triggered lamp, a trained photo sorter, a remote-controlled car, a calculator, an unspecified smart speaker and an unspecified robot vacuum.

Classify all six using **mechanism evidence**. Which two need **more information**?

The lamp uses a fixed rule; the photo sorter uses a **learned model**; the car follows human commands; the calculator performs arithmetic. The speaker and vacuum need **mechanism** details. Products can combine **mechanism**s.

A robot body or smart label proves none of them.

▼ **Reference: the original lesson route**

REFERENCE · SECTION 15

Reference: the original lesson route

> The main route at the start of this edition replaces the need to follow several overlapping plans.

Goals and evidence

By the end, I can	How I will show it
Distinguish an observed result from an explanation of how it was produced	Explain the deciding step, show a trace or test, and state one limit.
Identify written rules, learned models and human decisions inside one product	Explain the deciding step, show a trace or test, and state one limit.
Explain what additional evidence is needed when a mechanism is unknown	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Original umbrella opening question**

Start with this question: A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?

▼ **Optional supported practice**

REFERENCE · SECTION 17

Optional supported practice

> Choose the interactive questions or this paper version. They refer to the same practice, not two separate assignments.

Question 1: identify the deciding stage

- A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?
- Which labelled step matches this detail? The teacher checks current conditions before distributing umbrellas.
- 1. Weather records
- 2. Final decision
- 3. Display rule
- 4. Forecast

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check after your attempt**

After your attempt, compare with the supported explanation: Final decision The teacher checks current conditions before distributing umbrellas. The **fitting** record supports calling the forecasting stage learned.

Comparing 70 with 60 is an explicit rule. The final action belongs to a person. Merely seeing TAKE ONE would not reveal any of these **mechanism**s; the **process** notes supply that **evidence**.

Question 2: change the boundary

- Changed-case practice: Keep the forecast but raise the display boundary to **80%**. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. The display no longer says TAKE ONE.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after your attempt**

After your attempt, compare with the supported explanation: The display no longer says TAKE ONE. 70 is below 80; changing a display rule does not refit the model.

Keep the work you completed

This is the starting lesson. Ask your teacher to model one **observation**, one **written rule** and one deciding step, then try your own example.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Original practice and portfolio headings**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

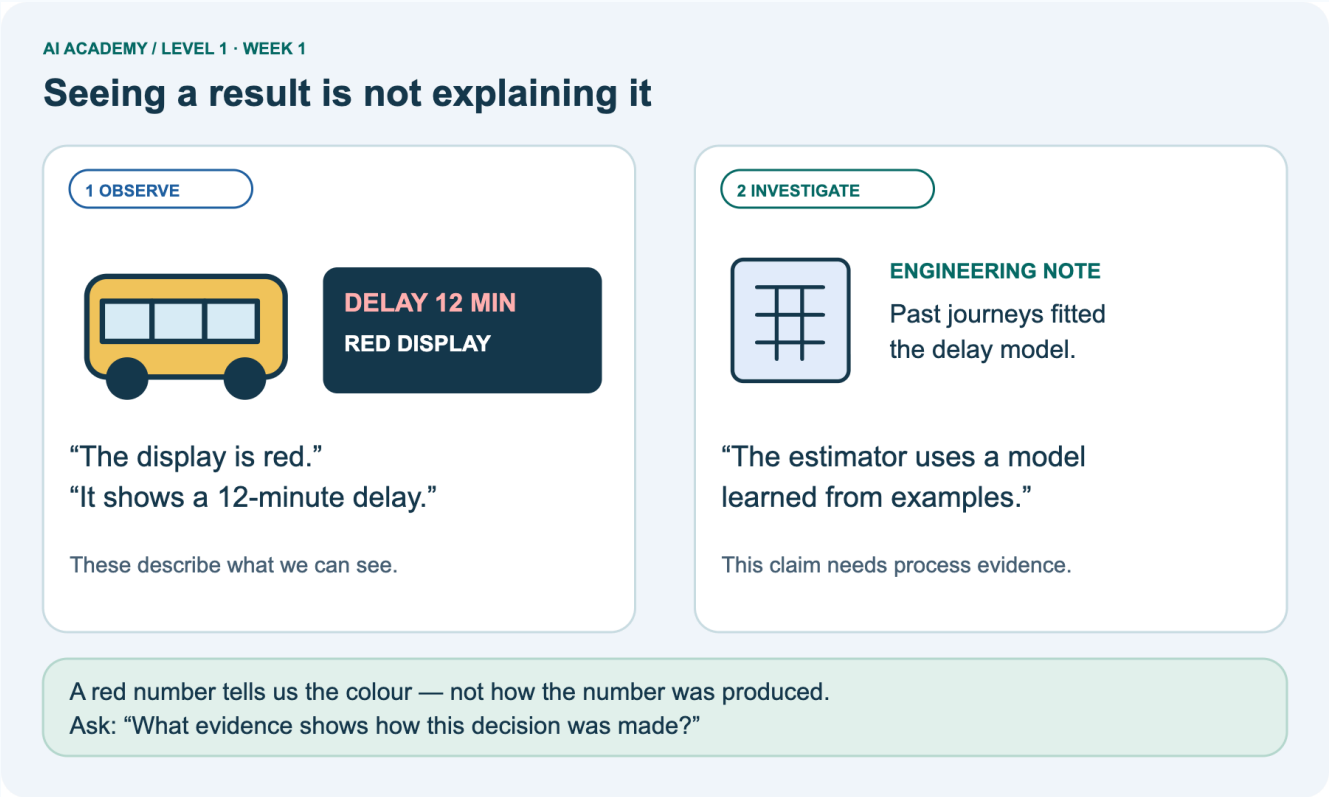
Keep your completed work

▼ **The six-picture reference gallery**

The six-picture reference gallery

> Revisit a picture when you need it. Key pictures also appear beside the main explanation.

01. Seeing a result is not explaining it

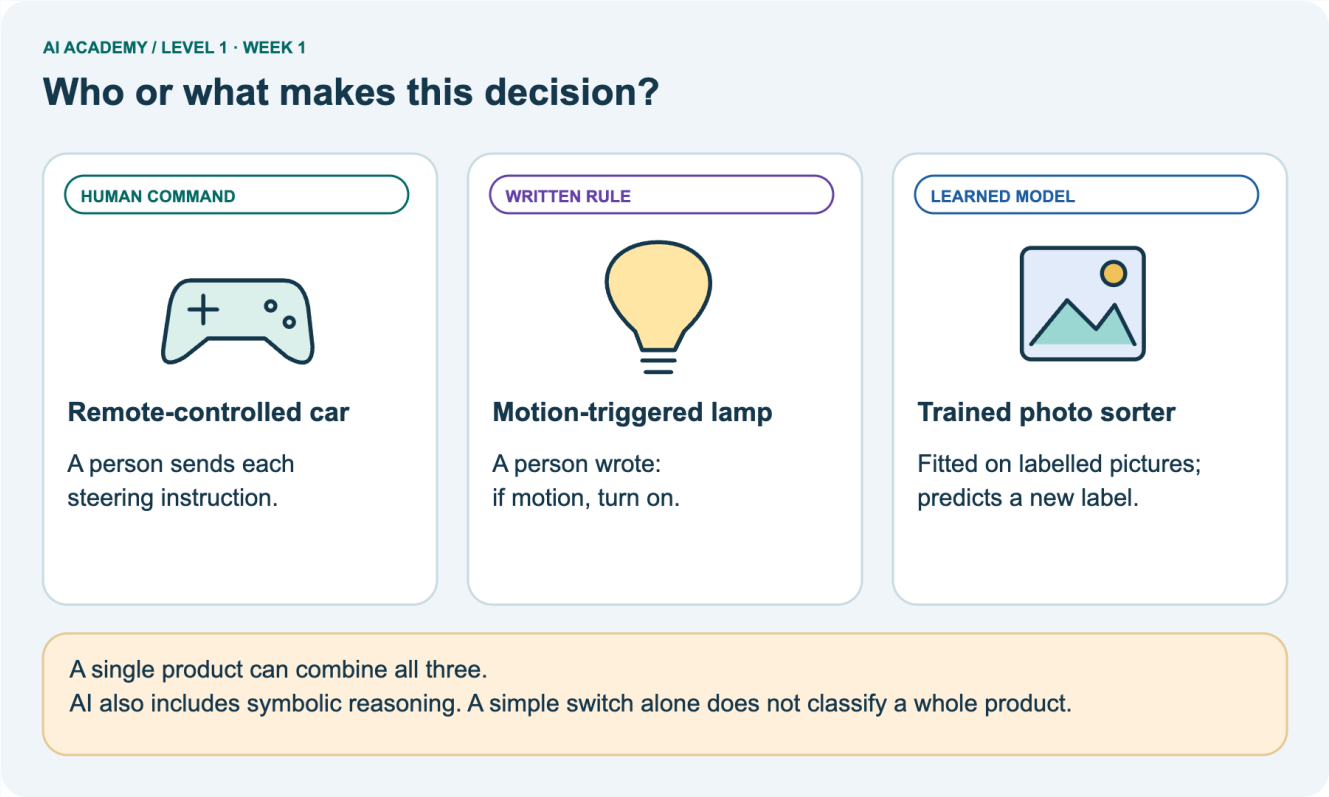


Follow the labelled picture with the matching explanation. Open the vector diagram

A bus has a red display showing a 12-minute delay. That is an **observation**. A supplied engineering note says past journeys fitted the delay model; that is **evidence** about its **mechanism**.

Name one **observation** and one claim that needs more **evidence**.

02. Who or what makes this decision?

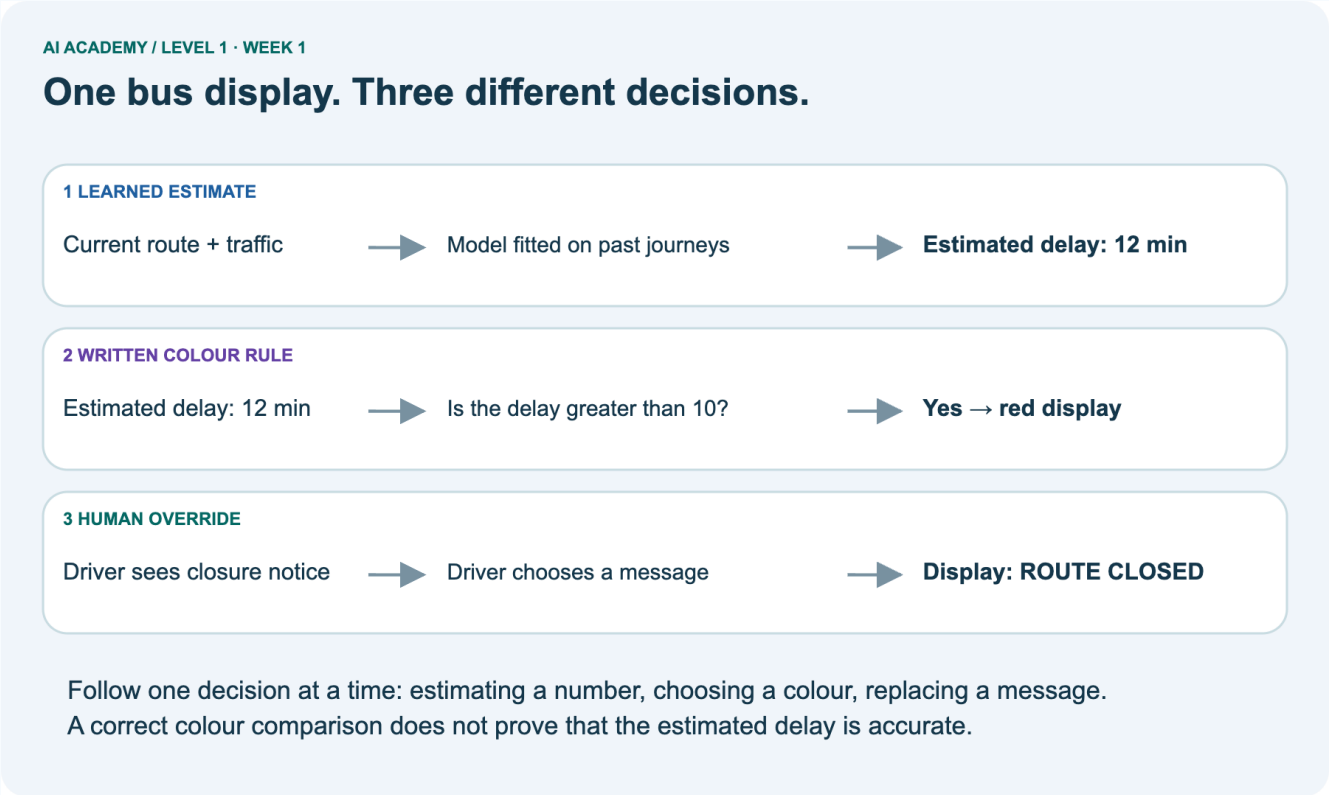


Follow the labelled picture with the matching explanation. Open the vector diagram

A remote-controlled car follows human commands, a motion lamp follows a **written rule**, and a trained photo sorter applies a **learned model**. A product can combine **mechanism**s. AI also includes symbolic reasoning.

Point to who or what makes each decision.

03. One bus display. Three different decisions.

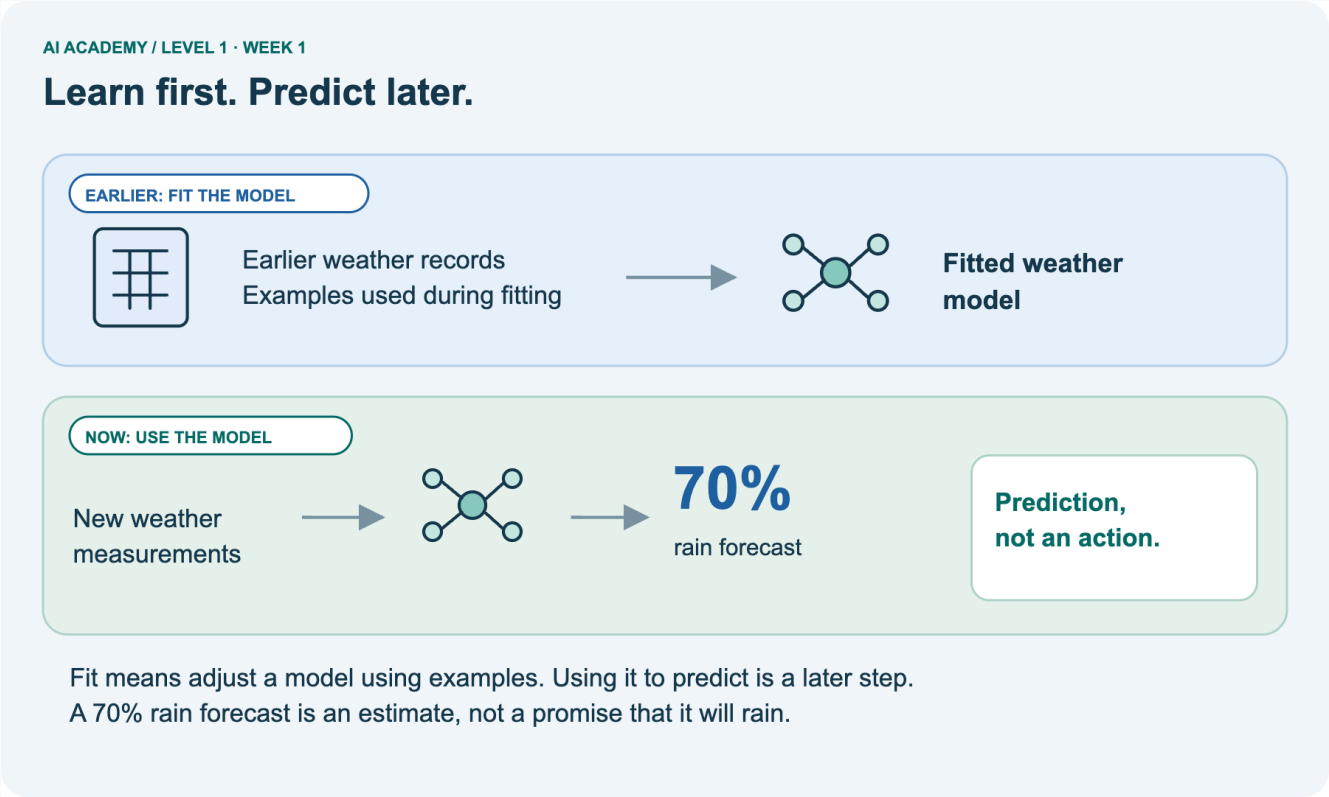


Follow the labelled picture with the matching explanation. Open the vector diagram

Three separate rows trace the bus estimate, display colour and driver override. The fitted model estimates **12 minutes**; the written greater-than-10 rule makes the display red; the driver can choose ROUTE CLOSED.

Trace the number, colour and override separately.

04. Learn first. Predict later.

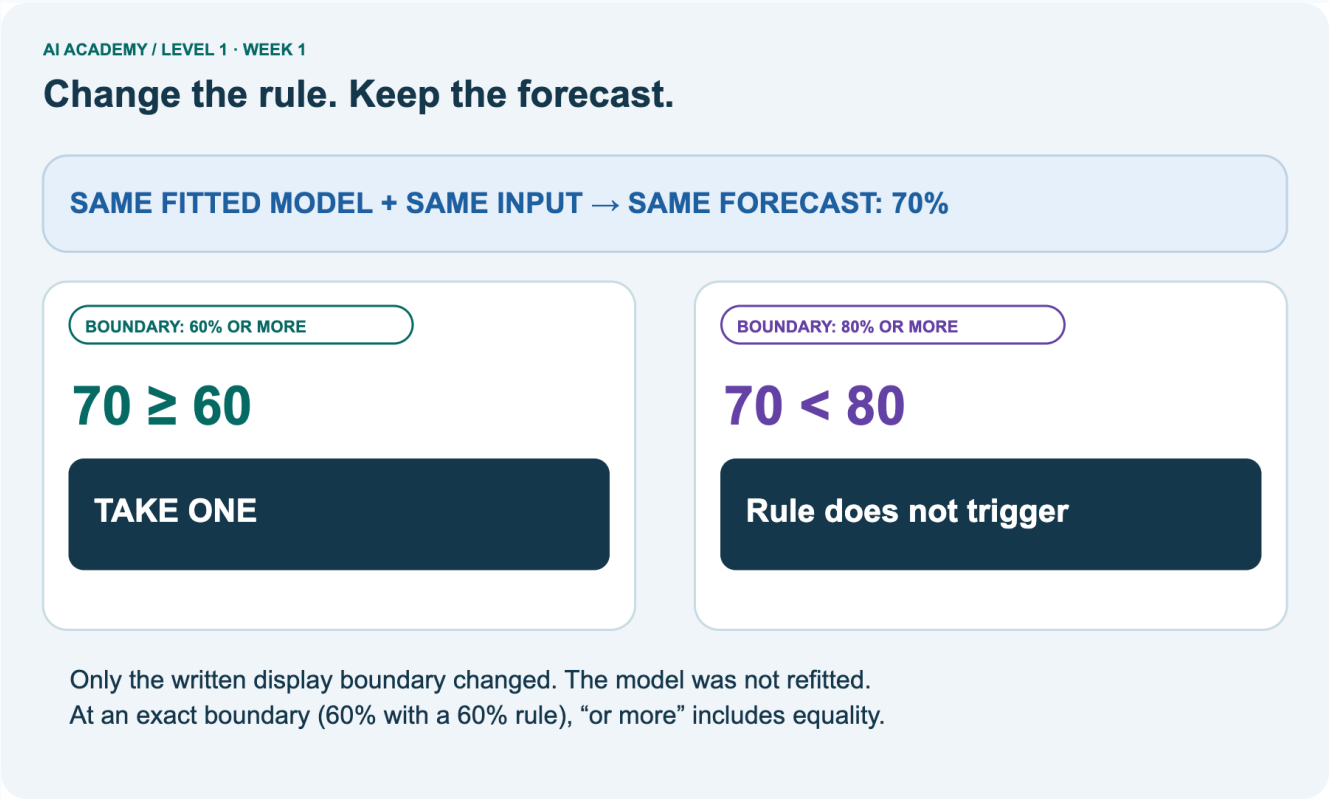


Follow the labelled picture with the matching explanation. Open the vector diagram

Earlier weather records are used to fit a model. Later, new measurements are passed to that fitted model to obtain a 70 percent rain forecast. It is an estimate, not a guarantee.

Which steps use earlier examples, and which use a fitted model?

05. Change the rule. Keep the forecast.

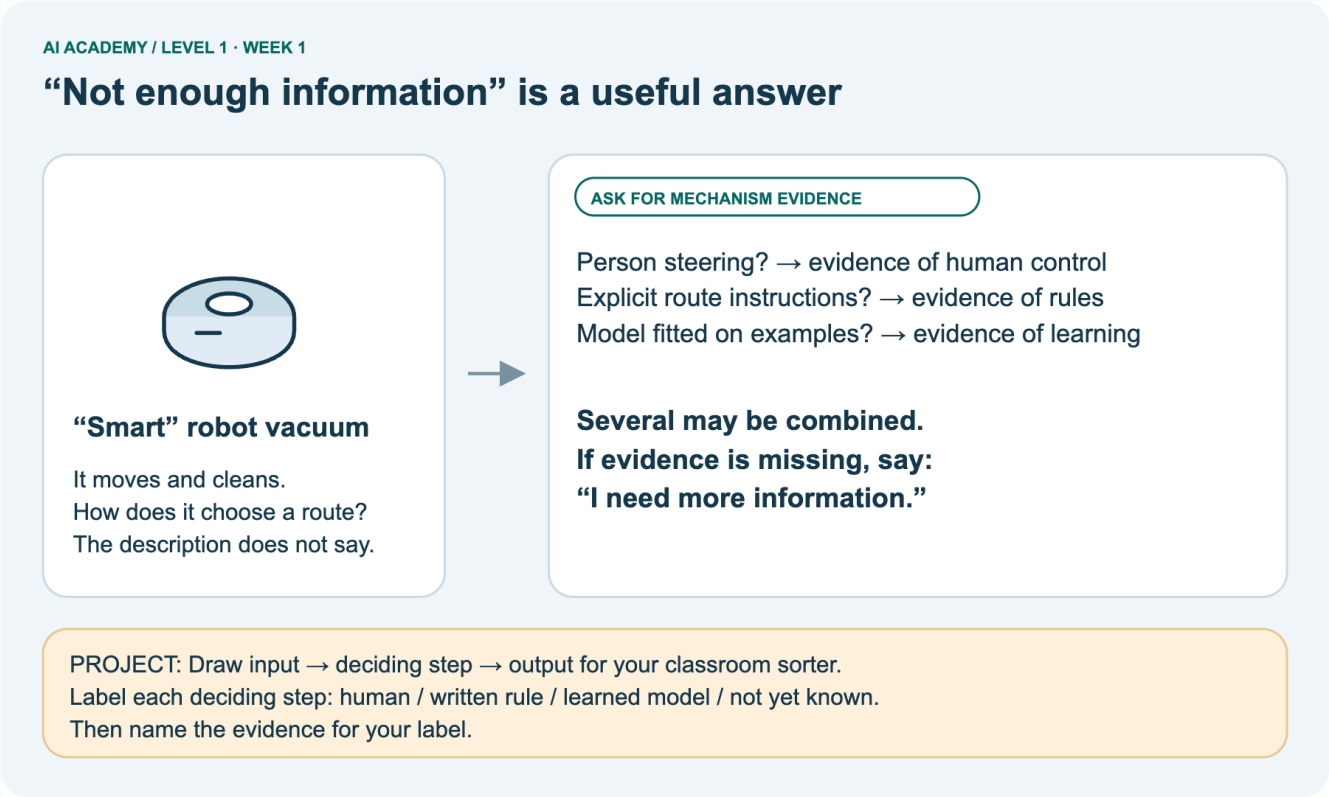


Follow the labelled picture with the matching explanation. Open the vector diagram

With a 70 percent rain forecast, a 60 percent-or-more rule displays TAKE ONE. An 80 percent-or-more rule does not trigger that message. Only the boundary changed; the fitted model and forecast did not.

Keep the forecast fixed at **70%**. What changes when the boundary becomes **80%**?

06. “Not enough information” is a useful answer



Follow the labelled picture with the matching explanation. Open the vector diagram

An unspecified smart robot vacuum moves and cleans, but its route **mechanism** is unknown. Ask for **evidence** of human commands, **written rule**s or model **fitting**, and allow combinations. Apply this reasoning to a classroom sorter.

Which **evidence** would change your answer about the vacuum?

▼ Original graphic headings

01. Seeing a result is not explaining it

02. Who or what makes this decision?

03. One bus display. Three different decisions.

04. Learn first. Predict later.

05. Change the rule. Keep the forecast.

06. “Not enough information” is a useful answer

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.